Problem 1: Linear functions

(a) Is $\cos x$ a linear function?

(b) Are polynomials of the form $f(x) = ax^2 + bx + c$ linear functions?

Problem 2: Dimension of a Span

Find the dimension of the span of each set of vectors in $\mathbb{R}^4$.

(a) $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 2 \\ 4 \\ 1 \\ 0 \end{bmatrix}$, $\vec{v}_3 = \begin{bmatrix} 3 \\ 6 \\ 1 \\ 1 \end{bmatrix}$, $\vec{v}_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -2 \end{bmatrix}$

(b) $\vec{w}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{w}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$, $\vec{w}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$, $\vec{w}_4 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

Problem 3: Rank

(a) Find the rank of

$$A = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \\ 1 & 2 & 2 & 4 \end{bmatrix}$$

(b) Let

$$C = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & c \end{bmatrix}$$

For which values of c does C have rank 3, and for which does it have rank 2?

Problem 4: Null Space

(a) Let A be a 5×7 matrix. What is the largest possible rank of A? What is the smallest dimension of the null space of A?

(b) True or false (give a short reason): if a 4×4 matrix has linearly independent columns, then its null space is $\{\vec{0}\}$ and its rows are also linearly independent.